

Continuum Mechanics

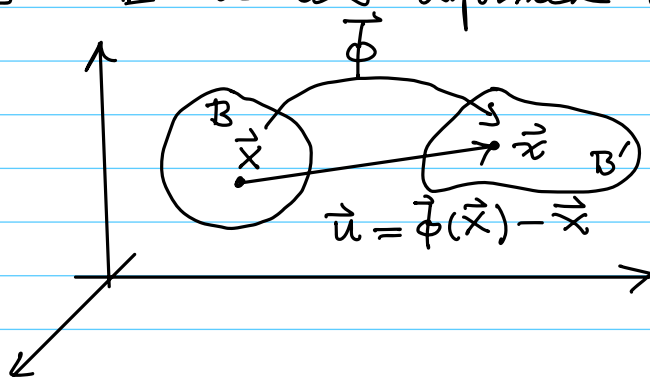
class # 12

9/24/2026

Chapter 4 Kinematics

Deformations

Definition: let $B \subseteq E^3$ be an undeformed configuration and $B' \subseteq E^3$ be its deformed configuration.



B is called the reference configuration and B' is called the deformed configuration.

We denote the points of B by

$$\vec{X} = (x_1, x_2, x_3)^T$$

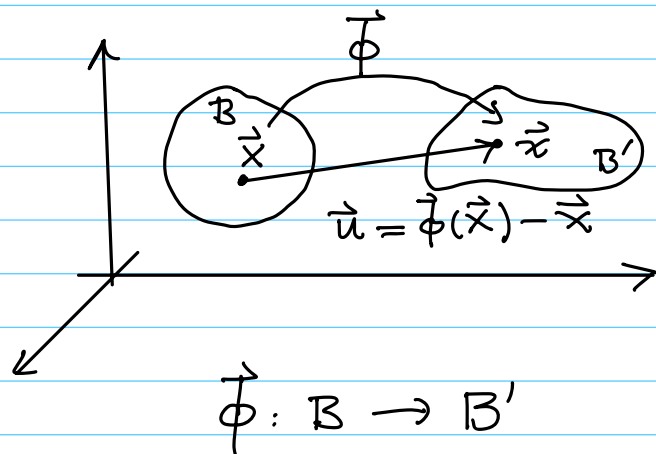
and points of B' are denoted by

$$\vec{x} = (x_1, x_2, x_3)^T.$$

B and B' represent characterizations of the same body, one before deformation and one after.

The coordinates X_i are called material coordinates, while x_i are called spatial coordinates.

Definition: The deformation of a body from a configuration B onto the configuration B' is described by a function



which maps each point $\vec{x} \in B$ to a point

$$\vec{\bar{x}} = \vec{\phi}(\vec{x}) \in B'.$$

We call $\vec{\phi}$ the deformation map relative to the reference configuration B .

The displacement of a material point from its location $\vec{x}$ to its final location $\vec{\bar{x}} = \vec{\phi}(\vec{x})$ is given by

$$(3.1) \quad \vec{u}(\vec{x}) = \vec{\phi}(\vec{x}) - \vec{x}. \quad //$$

Of course $\vec{u}$ is a function

$$\vec{u}: B \rightarrow V.$$

Definition: Let $B \subseteq \mathbb{E}^3$ be a reference configuration and suppose

$$\vec{\phi}: B \rightarrow B'$$

is its deformation map. $\vec{\phi}$ is called admissible

iff

i) $\phi: B \rightarrow B'$ is one-to-one and onto,

ii) $\det \nabla \vec{\phi}(\vec{X}) > 0, \forall \vec{X} \in B.$ //

Remark: Assumption (ii) implies that admissible deformations preserve the self-orientation of a body. In particular, a body cannot be continuously deformed onto its mirror image.

Deformation Gradient

Definition: Suppose that $\vec{\phi}: B \rightarrow B'$ is an admissible deformation map that is smooth. The deformation gradient $F: B \rightarrow \mathcal{V}^2$ is defined as

$$F(\vec{X}) = \nabla \vec{\phi}(\vec{X}). //$$

The tensor field F characterizes the local behavior of a deformation. Recall, from Taylor's Theorem

$$\vec{\phi}(\vec{X}) = \vec{\phi}(\vec{X}_0) + F(\vec{X}_0)(\vec{X} - \vec{X}_0) + O(|\vec{X} - \vec{X}_0|^2)$$

Equivalently,

$$(4.2) \quad \vec{\phi}(\vec{X}) = \vec{c} + F(\vec{X}_0)\vec{X} + O(|\vec{X} - \vec{X}_0|^2),$$

where

$$\vec{c} := \vec{\phi}(\vec{X}_0) - F(\vec{X}_0)\vec{X}_0$$

Thus, $F(\vec{X}_0)$ characterizes the local behavior of $\vec{\Phi}$ around a point $\vec{X}_0$.

Definition: A deformation $\vec{\Phi}: B \rightarrow B'$ is called homogeneous iff $F(\vec{X}) = F_0$, for all $\vec{X} \in B$.

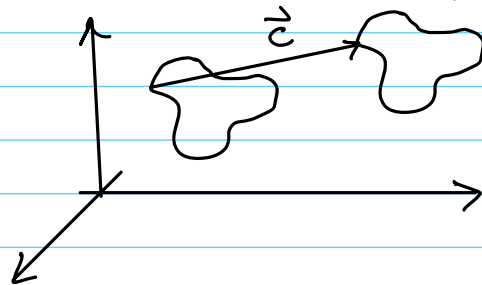
In this case, the approximation in (4.2) becomes exact:

$$(4.3) \quad \phi(\vec{X}) = \vec{c} + F\vec{X}, \quad \forall \vec{X} \in B.$$

Definition: A deformation $\vec{\Phi}$ is called a translation iff

$$\vec{\Phi}(\vec{X}) = \vec{X} + \vec{c}. //$$

In this case, $F = I$. There is no change in the shape or orientation of the body.



$$\vec{u} = \vec{\Phi}(\vec{X}) - \vec{X} = \vec{c}, \quad \forall \vec{X} \in B.$$

This deformation is clearly homogeneous.

Definition: Suppose $\vec{\phi}: B \rightarrow B'$ is a homogeneous deformation, i.e.,

$$\vec{\phi}(\vec{x}) = \vec{c} + F\vec{x}$$

for some $F \in \mathcal{V}^2$. $\vec{\phi}$ is said to have a fixed point at $\vec{y} \in B$ iff

$$(4.4) \quad \vec{\phi}(\vec{x}) = \vec{y} + F(\vec{x} - \vec{y}), \quad \forall \vec{x} \in B.$$

In this case

$$\vec{\phi}(\vec{y}) = \vec{y}.$$

Result (4.1): let $\vec{\phi}$ be an arbitrary homogeneous deformation with constant deformation gradient $F \in \mathcal{V}^2$. Then, given any point $\vec{y}$, we can decompose $\vec{\phi}$ as

$$\vec{\phi} = \vec{d}_1 \circ \vec{g} = \vec{g} \circ \vec{d}_2,$$

where $\vec{d}_1$ and $\vec{d}_2$ are translations and $\vec{g}$ is homogeneous with a fixed point $\vec{y}$. In particular,

$$\vec{g}(\vec{x}) = \vec{y} + F(\vec{x} - \vec{y}).$$

Proof: Exercise. ///

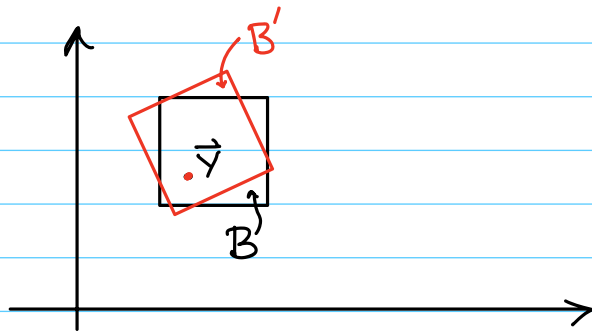
Rotations and Stretches

Definition: A homogeneous deformation $\vec{\phi}$ is called a rotation about a fixed point $\vec{Y}$ iff

$$\vec{\phi}(\vec{x}) = \vec{Y} + Q(\vec{x} - \vec{Y})$$

and Q is a rotation tensor, that is,

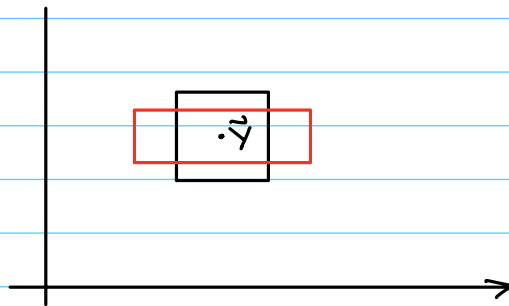
$$Q^T Q = I \text{ and } \det(Q) = 1$$



Definition: A homogeneous deformation $\vec{\phi}$ is called a stretch from $\vec{Y}$ iff

$$\vec{\phi}(\vec{x}) = \vec{Y} + S(\vec{x} - \vec{Y}),$$

where S is a symmetric positive definite tensor.



Example: Suppose that $\vec{Y} = (2, 2, 2)^T$

$$B = \{(x_1, x_2, x_3)^T \in \mathbb{E}^3 \mid x_1, x_2, x_3 \in (1, 3)\}$$

Suppose that

$$S = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

set

$$\vec{\phi}(\vec{x}) = \vec{Y} + S(\vec{x} - \vec{Y}).$$

Then $B' = \vec{\phi}(B)$ is given by

$$B' = \{(x_1, x_2, x_3)^T \in \mathbb{E}^3 \mid x_1 \in (2, 4), x_2 \in (3/2, 5/2), x_3 \in (1, 3)\}$$

Result (4.2): let $\vec{\phi} : B \rightarrow B'$ be a homogeneous deformation with a fixed point $\vec{Y}$, that is,

$$(4.4) \quad \phi(\vec{x}) = \vec{Y} + F(\vec{x} - \vec{Y})$$

for some $F \in \mathcal{V}^2$. let

$$F = RU = VR$$

be the right and left polar decompositions given in Result 1.12. Then $\vec{\phi}$ can be decomposed as

$$\vec{\phi} = \vec{r} \circ \vec{S}_1 = \vec{S}_1 \circ \vec{r},$$

where

$$\vec{r}(\vec{x}) = \vec{y} - R(\vec{x} - \vec{y})$$

is a rotation about $\vec{y}$ and

$$\vec{s}_1(\vec{x}) = \vec{y} + U(\vec{x} - \vec{y})$$

and

$$\vec{s}_2(\vec{x}) = \vec{y} + V(\vec{x} - \vec{y})$$

are stretches from $\vec{y}$.

Proof: Exercise. ///

Extensions

Definition: A homogeneous deformation $\vec{\Phi}$ is called an extension at $\vec{y}$ in the unit direction $\vec{e}$, $|\vec{e}| = 1$,

iff

$$\vec{\Phi}(\vec{x}) = \vec{y} + F(\vec{x} - \vec{y})$$

and

$$F = I + (\lambda - I)\vec{e} \otimes \vec{e}$$

for some $\lambda > 0$.

Remark: F extends any vector parallel to e by a factor of $\lambda > 0$. Clearly,

$$F\vec{e} = \vec{e} + (\lambda - 1)(\vec{e} \cdot \vec{e})\vec{e} = \lambda\vec{e}$$

Proposition: An extension deformation is a stretch deformation.

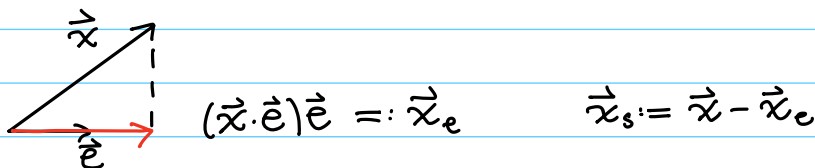
Proof: We only need to prove that F defined above is SPD. So

$$F = I + (\lambda - I) \vec{e} \otimes \vec{e},$$

for some $\vec{e} \in V$.

$$\begin{aligned} \vec{x} \cdot (F \vec{x}) &= \vec{x} \cdot \left(\vec{x} + (\lambda - I)(\vec{e} \cdot \vec{x}) \vec{e} \right) \\ &= \vec{x} \cdot \vec{x} + (\lambda - I)(\vec{e} \cdot \vec{x})^2 \\ &= |\vec{x}|^2 + \lambda(\vec{e} \cdot \vec{x})^2 - (\vec{e} \cdot \vec{x})^2 \end{aligned}$$

Now, suppose $\vec{x} \cdot \vec{e} \neq 0$. Then,



By the Pythagorean Theorem

$$\begin{aligned} |\vec{x}|^2 &= |(\vec{x} \cdot \vec{e}) \vec{e}|^2 + |\vec{x}_s|^2 \\ &\geq |(\vec{x} \cdot \vec{e}) \vec{e}|^2 \\ &= (\vec{e} \cdot \vec{x})^2 \end{aligned}$$

Thus

$$|\vec{x}|^2 - (\vec{e} \cdot \vec{x})^2 \geq 0$$

So

$$\vec{x} \cdot (F \vec{x}) \geq \lambda (\vec{e} \cdot \vec{x})^2 > 0.$$

On the other hand, if $\vec{x} \cdot \vec{e} = 0$, but $\vec{x} \neq 0$,

$$\vec{x} \cdot F\vec{x} = |\vec{x}|^2 > 0.$$

Thus F is positive definite. It is also clearly symmetric. Thus, F is SPD. ///

Result 4.3 (Stretches as Extensions): Let $\vec{S}_1$ and $\vec{S}_2$ be the stretches defined in Result 4.2, and let $\{\lambda_i, \vec{u}_i\}$ and $\{\lambda_i, \vec{v}_i\}$, $i=1,2,3$ be the eigenpairs associated with U and V , respectively. Then,

$$\vec{S}_1 = \vec{f}_1 \circ \vec{f}_2 \circ \vec{f}_3 \quad \vec{S}_2 = \vec{h}_1 \circ \vec{h}_2 \circ \vec{h}_3,$$

where $\vec{f}_i$ is an extension at λ_i in the direction $\vec{u}_i$ and $\vec{h}_i$ is an extension at λ_i in the direction $\vec{v}_i$.

Proof: Recall

$$\phi(\vec{X}) = \vec{Y} + F(\vec{X} - \vec{Y})$$

and

$$F = RU = VR$$

where

$$U = \sqrt{F^T F} \quad V = \sqrt{F F^T}$$

are SPD matrices and R is a rotation, i.e.,

$$R^T R = I, \quad \det(R) = 1.$$

It is a fact that U and V have the same eigenvalues, and

$$\lambda_1, \lambda_2, \lambda_3 > 0,$$

provided that F is invertible with positive determinant. In fact, this must be the case.

By the spectral decomposition Theorem,

$$U = \tilde{U} \Delta \tilde{U}^T = \sum_{i=1}^3 \lambda_i \vec{u}_i \otimes \vec{u}_i$$

where

$$\Delta = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

$$\tilde{U} = \begin{bmatrix} 1 & 1 & 1 \\ \vec{u}_1 & \vec{u}_2 & \vec{u}_3 \\ 1 & 1 & 1 \end{bmatrix}, \quad \tilde{U} \tilde{U}^T = \mathbf{I},$$

and similarly for V . By Result 4.2

$$\vec{\phi}(\vec{x}) = \vec{r}(\vec{z}_1(\vec{x}))$$

and

$$\begin{aligned} \vec{z}_1(\vec{x}) &= \vec{y} + U(\vec{x} - \vec{y}) \\ &= \vec{y} + \tilde{U} \Delta \tilde{U}^T (\vec{x} - \vec{y}) \end{aligned}$$

Now write

$$\vec{f}_i(\vec{x}) := \vec{y} + F_i(\vec{x} - \vec{y})$$

where

$$F_i = I + (\lambda_i - 1) \vec{u}_i \otimes \vec{u}_i \quad (\text{no sum})$$

The key is to use the decomposition

$$U = \sum_{i=1}^3 \lambda_i \vec{u}_i \otimes \vec{u}_i$$

Then, it can be shown that

$$\begin{aligned} (\vec{f}_2 \circ \vec{f}_3)(\vec{x}) &= \vec{f}_2(\vec{f}_3(\vec{x})) \\ &= \vec{y} + F_2 F_3 (\vec{x} - \vec{y}) \end{aligned}$$

and

$$(\vec{f}_1 \circ \vec{f}_2 \circ \vec{f}_3)(\vec{x}) = \vec{y} + F_1 F_2 F_3 (\vec{x} - \vec{y})$$

Furthermore,

$$F_1 F_2 F_3 = I + \sum_{i=1}^3 (\lambda_i - 1) \vec{u}_i \otimes \vec{u}_i.$$

One can also show that, since $\{\vec{u}_1, \vec{u}_2, \vec{u}_3\}$ form an orthonormal basis for V ,

$$I = \sum_{i=1}^3 \vec{u}_i \otimes \vec{u}_i.$$

Hence

$$\begin{aligned} F_1 F_2 F_3 &= \sum_{i=1}^3 \vec{u}_i \otimes \vec{u}_i + \sum_{i=1}^3 (\lambda_i - 1) \vec{u}_i \otimes \vec{u}_i \\ &= \sum_{i=1}^3 \lambda_i \vec{u}_i \otimes \vec{u}_i \\ &= U. \end{aligned}$$

We have already shown that

$$(\vec{f}_1 \circ \vec{f}_2 \circ \vec{f}_3)(\vec{x}) = \vec{y} + F_1 F_2 F_3 (\vec{x} - \vec{y})$$

and

$$F_1 F_2 F_3 = U.$$

Thus

$$(\vec{f}_1 \circ \vec{f}_2 \circ \vec{f}_3)(\vec{x}) = \vec{S}_1(\vec{x}).$$

The result for $\vec{S}_2$ is proven in the same way. ///